

Early Postoperative Gastrointestinal Tract Evaluation: Expected Findings and Complications

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Postoperative imaging evaluation following gastrointestinal surgery is a critical concern for radiologists and surgeons. The early postoperative period can be challenging for radiologists because the imaging appearances of expected findings and complications are often similar. The slide presentation highlights the importance of correctly identifying imaging findings that require further investigation.

Free intraperitoneal air and fluid, fat stranding, and bowel dilatation are commonly expected in the early postoperative period. However, these findings can also serve as indicators of postoperative complications. Understanding the specific imaging characteristics and their evolution over time is crucial for distinguishing expected findings from potential complications.

Contrast-enhanced CT is the most commonly used imaging modality for evaluation of postoperative patients. The CT protocol must be individually tailored to each patient based on the surgical procedure performed and the clinical suspicions. For example, administration of positive oral contrast agent aids in detection of complications such as anastomotic leakage (Fig 1). Use of the appropriate CT protocol ensures early detection of complications and leads to timely intervention.

The slide presentation enhances the reader's abilities in the following areas: (a) differentiating between expected postsurgical imaging findings and surgical complications in the early postoperative period, (b) selecting the appropriate CT protocol for assessing different suspected complications in the early postoperative period, and (c) identifying specific imaging patterns associated with intraperitoneal air, intraperitoneal fluid, fat stranding, and bowel dilatation that will aid in diagnosis.

By understanding the expected postoperative findings and commonly seen complications and by effectively performing contrast-enhanced CT, radiologists can confidently evaluate patients after gastrointestinal surgery, promptly identify problems such as active bleeding (Fig 2), and contribute to optimized patient care and outcomes.



Figure 1. Coronal contrast-enhanced CT image obtained with positive oral contrast agent administration in a patient 10 days after pancreaticoduodenectomy shows leakage of intraluminal contrast material and fistulas (arrowheads) extending from the gastroenteric anastomosis (arrows) to a collection (*) in the subcutaneous tissue adjacent to the surgical wound in the anterior abdominal wall.

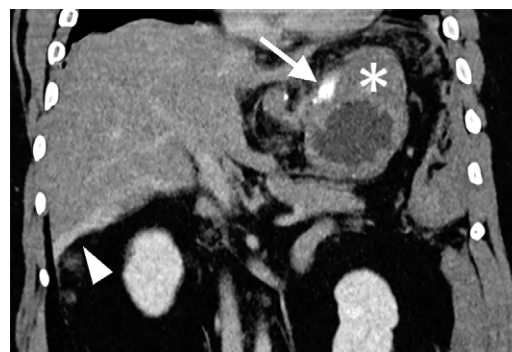


Figure 2. Coronal contrast-enhanced CT image in a patient 7 days after gastric bypass surgery shows a hyperattenuating collection (*) adjacent to the bypassed stomach, with contrast blooming (arrow) inside the perigastric collection and enhancing subhepatic fluid (arrowhead). The findings are indicative of active bleeding.

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TEACHING POINTS

- Contrast-enhanced CT is a commonly used imaging modality for evaluating postoperative patients because it accurately characterizes gas and fluid collections. Administration of positive oral and rectal contrast material can be beneficial in patients suspected of having anastomotic leakage or bowel perforations.
- Free intraperitoneal air and fluid, fat stranding, and bowel dilatation are expected in the early postoperative period. However, these findings can also serve as indicators of postoperative complications.
- Unexpected postsurgical imaging findings include venous thrombosis, peripheral wedge-shaped hypoenhancement, and foreign body reaction.

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Suggested Readings

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